

Customer Churn Prediction System

Machine Learning Internship Project

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Task ID: ML-2

1. Introduction

Customer churn refers to customers discontinuing the use of a company's products or services. Predicting churn helps businesses proactively retain customers, improve customer satisfaction, and increase revenue.

The objective of this project is to develop a machine learning system capable of predicting whether a customer is likely to churn based on demographic information, spending habits, and engagement patterns.

2. Dataset Description

A synthetic dataset containing **2,000 customer records** was created to simulate a subscription-based business.

The dataset contains the following features:

Feature	Description
CustomerID	Unique customer identifier
Age	Customer age
Gender	Male/Female
City	Customer city
SubscriptionType	Basic, Premium, Standard
MonthlySpending	Monthly customer spending
Tenure	Number of months as customer
NumberOfPurchases	Total purchases
SupportRequests	Customer support requests
LoginFrequency	Number of logins
LastActivityDays	Days since last activity
SatisfactionScore	Customer satisfaction (1–10)
Churn	Target variable

The dataset was generated using Python with realistic business rules and randomized customer behavior.

3. Data Preparation

Several preprocessing steps were performed before model training.

- Checked for missing values.
 - Removed duplicate records.
 - Encoded categorical variables using LabelEncoder.
 - Created new engineered features.
 - Verified data consistency.
 - Split dataset into training and testing sets using an 80:20 ratio.
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4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to better understand customer behavior.

The following visualizations were created:

- Churn Distribution
- Age Distribution
- Monthly Spending Distribution
- Correlation Heatmap
- Customer Satisfaction Distribution

Observations

- Customers with lower satisfaction scores were more likely to churn.
 - Customers with longer inactivity periods showed higher churn probability.
 - Premium customers generally had lower churn rates.
 - Customers with more support requests were at greater risk of leaving.
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5. Feature Engineering

Three additional features were created to improve model performance.

AveragePurchase

$\text{AveragePurchase} = \text{MonthlySpending} / (\text{NumberOfPurchases} + 1)$

This estimates the customer's average spending per purchase.

EngagementScore

$\text{EngagementScore} = \text{LoginFrequency} \times \text{SatisfactionScore}$

This represents overall customer engagement.

SupportPerMonth

$\text{SupportPerMonth} = \text{SupportRequests} / (\text{Tenure} + 1)$

This measures how frequently customers require support relative to their tenure.

These engineered features helped the model capture customer behavior more effectively.

6. Model Selection

Two classification models were trained and compared.

Random Forest

Random Forest is a robust ensemble learning algorithm that performs well on tabular datasets and provides strong baseline performance.

XGBoost

XGBoost is a gradient boosting algorithm known for its high predictive accuracy, speed, and ability to model complex relationships within structured data.

7. Model Evaluation

The models were evaluated using several performance metrics.

Random Forest

Accuracy: 98.75%

Precision: 98.43%

Recall: 97.66%

F1 Score: 98.04%

ROC-AUC: 99.82%

XGBoost

Accuracy: 99.25%

Precision: 99.21%

Recall: 98.44%

F1 Score: 98.82%

ROC-AUC: 99.89%

Selected Model

XGBoost was selected as the final model because it achieved the highest overall performance across all evaluation metrics.

The trained model was saved using Joblib and integrated into the Streamlit application.

8. Prediction Interface

A Streamlit-based web application was developed.

The application allows users to:

- Enter customer information.
- Predict customer churn.
- Display churn probability.
- Show customer risk level.
- Display feature importance.

- Explain predictions using SHAP Explainable AI.

This makes the model interactive and suitable for business users.

9. Key Findings

The following insights were observed:

- Customer satisfaction is one of the strongest predictors of churn.
 - High support requests significantly increase churn risk.
 - Long inactivity periods are associated with customer loss.
 - Longer customer tenure generally reduces churn probability.
 - Higher engagement leads to improved customer retention.
 - Monthly spending also influences customer loyalty.
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10. Business Recommendations

Based on the analysis, the following recommendations are suggested:

1. Identify high-risk customers early using the prediction system.
2. Improve customer support response times.
3. Offer loyalty rewards for long-term customers.
4. Increase customer engagement through personalized notifications.
5. Provide targeted discounts to customers with high churn probability.
6. Regularly monitor customer satisfaction scores.

These strategies can reduce churn and improve customer retention.

11. Conclusion

This project successfully developed a Customer Churn Prediction System using Machine Learning.

The solution includes data generation, preprocessing, exploratory analysis, feature engineering, model training, evaluation, and deployment using Streamlit.

Among the evaluated models, XGBoost achieved the highest performance with approximately **99.25% accuracy**.

The Streamlit application enables users to predict customer churn, visualize prediction confidence, and understand the factors influencing the prediction through SHAP Explainable AI.

Overall, the project demonstrates a complete end-to-end machine learning workflow suitable for real-world customer retention applications.